

ST CATHERINE'S SCHOOL

YEAR 12 - 4 UNIT (ADDITIONAL) MATHEMATICS
TIME ALLOWED: 3 HOURS (*plus 5 mins reading time*)

DATE: AUGUST, 1997

Student Number: 55

INSTRUCTIONS:

- All questions are to be attempted.
- All questions are of equal value.
- All necessary working should be shown in every question, as part of your solution.
- Full marks may not be awarded for careless or badly arranged work.
- Approved calculators and geometrical instruments are required.
- Standard Integrals are printed on the last page.
- Each question should be started in a *separate* Writing Booklet, clearly marked with the question number and your student number on the cover.
- You may ask for extra Writing Booklets if you need them.
- Tie your Booklets in 2 bundles (no staples are to be used):

Section A: Questions 1, 2, 3 and 4.
Section B: Questions 5, 6, 7 and 8.

- Hand in Section A, Section B and this examination paper separately

TEACHERS USE ONLY TOTAL MARKS
A
B
TOTAL

SECTION A

Question 1 (Use a separate Writing Booklet)

Marks

a) Find the following integrals:

4

i) $\int \sin 2x \cos x \, dx$

ii) $\int \sin 2x \cos 2x \, dx$

b) Show that $\int_4^5 \frac{2t^2 dt}{(t-1)(t-2)(t-3)} = 19 \ln 2 - 9 \ln 3$

3

c) Using integration by parts, or otherwise, find $\int e^x \sin 2x \, dx$

3

d) Show that $\int_0^m \frac{(m-x)^2}{m^2 + x^2} \, dx = m(1 - \ln 2)$

2

a) If $z_1 = 1 + 3i$, $z_2 = 1 - i$,

4

i) Find in the form $a + ib$, where a and b are real, the numbers $z_1 z_2$ and $\frac{z_1}{z_2}$.

ii) On an Argand Diagram the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ represent the complex numbers

$z_1 z_2$ and $\frac{z_1}{z_2}$ respectively (where z_1 and z_2 are given above).

Show this on an Argand Diagram, giving the co-ordinates of A and B .

iii) From your diagram, deduce that $\frac{z_1}{z_2} - z_1 z_2$ is real.

b) Given that $z = \sqrt{3} - i$,

4

i) Express z in modulus-argument form.

ii) Hence, evaluate the following in the form $x + iy$:

(α) z^5

(β) $(\bar{z})^5$

(γ) $\frac{z^5}{(\bar{z})^5}$

c) On an Argand diagram, sketch the locus of z if:

4

i) $|z + 3| < |z - 1 - 4i|$

ii) $4 \arg \frac{z-1}{z+3} = \pi$

Question 3 (Use a separate Writing Booklet)

Marks

- a) i) Show that $2 + i$ is a root of $2z^3 - 5z^2 - 2z + 15 = 0$ 3
 ii) Find the other roots.
- b) If the roots of the equation $x^3 - px^2 + qx - r = 0$ are α , β and γ : 4
 i) Find the equation with roots $\alpha + \beta$, $\beta + \gamma$, $\gamma + \alpha$.
 ii) Hence, or otherwise, find the value of $(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha)$
- c) i) Show that if the equation $P(x) = 0$ has a root of multiplicity m , then the equation $P'(x)$ has a root of multiplicity $(m-1)$. 5
 ii) Solve the equation $x^4 - x^3 - 9x^2 - 11x - 4 = 0$, given that it has a root of multiplicity 3.

Question 4 (Use a separate Writing Booklet)

- a) i) By expanding $(\cos \theta + i \sin \theta)^5$ in two different ways, 6
 obtain an expression for $\cos 5\theta$ in terms of powers of $\cos \theta$.
 ii) Hence solve the equation $16x^4 - 20x^2 + 5 = 0$ giving solutions in the form $x = \cos \alpha$.
- b) If $I_n = \int \frac{x^n}{\sqrt{x^2 - a^2}} dx$ 6
 i) Show that $nI_n - (n-1)a^2I_{n-2} = x^{n-1}\sqrt{x^2 - a^2}$
 ii) Hence evaluate $\int_2^4 \frac{x^4}{\sqrt{x^2 - 4}} dx$

SECTION B

Question 5 (Use a separate Writing Booklet)

Marks

- a) For the curve $y = \frac{x^2}{x^3 + 4}$ 7
- i) Find any horizontal or vertical asymptotes.
- ii) Find any maximum or minimum turning points.
- iii) Sketch the curve.
- iv) Use the graph to show that there are three solutions to the equation $x^3 - 4x^2 + 4 = 0$
-
- b) On a new set of axes sketch $y = \left| \frac{x^2}{x^3 + 4} \right|$ 2
-
- c) Find the domain of $y = \pm \sqrt{\frac{x^2}{x^3 + 4}}$ 1
-
- d) Without further use of calculus, sketch $y^2 = \frac{x^2}{x^3 + 4}$ 2

Question 6 (Use a separate Writing Booklet)

Marks

- a) Find $\int \cos^2 y \, dy$. 1
- b) A hyperbola has asymptotes $y = x$ and $y = -x$. It passes through the point $(3, 2)$. 4
- i) Find the equation of the hyperbola.
- ii) Determine its eccentricity and foci.
- c) Find the equation of the tangents to the ellipse $\frac{x^2}{27} + \frac{y^2}{9} = 1$ 3
- which are perpendicular to the line $x + y = 10$.
- d) For the hyperbola $xy = 16$, P is the variable point $\left(4p, \frac{4}{p}\right)$. 4
- i) Find the equations of the tangent and normal to the hyperbola at the point P .
- ii) If the tangent intersects the X axis at A and the normal intersects the Y axis at B , find the area of $\triangle PAB$.

- a) The region bounded by $y = x^2 + 2$, the x axis, the y axis and the line $x = 2$ is rotated about the line $x = 2$.

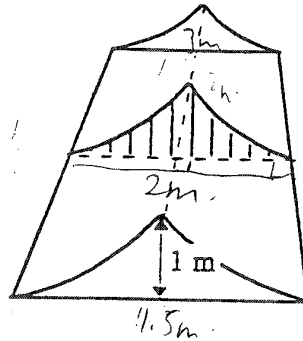
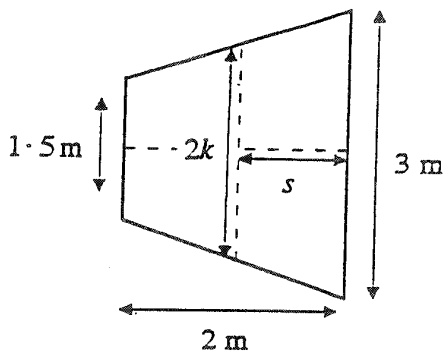
4

Use cylindrical shells to find the volume generated.

- b) Trapezium base of the tent

Tent showing typical cross-section

8



The base of a tent is a trapezium with parallel sides of length 1.5 metres at the back of the tent and 3 metres at the front of the tent. The base has an axis of symmetry perpendicular to the parallel sides and 2 metres long. The roof of the tent is formed by draping material over a horizontal ridge pole of length 2 metres directly above the axis of symmetry of the base and at a height of 1 metre, as shown in the diagram above.

A vertical cross-section taken perpendicular to the axis of symmetry of the base has the shape of the region shaded above and has area $\frac{1}{3}k$ square units, where $2k$ metres is the width of the cross-section where it meets the trapezium base.

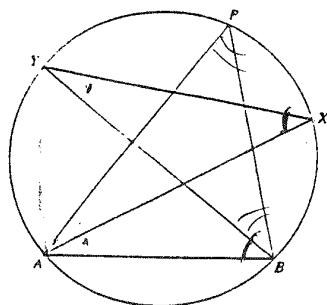
- Show that if at a distance s metres from the front of the tent (measured along the axis of symmetry of the trapezium) the width of the trapezium base is $2k$ metres, as shown in the diagram, then $k = \frac{3}{2}\left(1 - \frac{1}{4}s\right)$.
- Deduce that the area of typical cross-section as shaded above, taken at a distance s metres from the front of the tent, is $\frac{1}{2}\left(1 - \frac{1}{4}s\right)$ square units.
- If the tent has vertical flaps front and back, calculate the volume of the interior of the tent.

Question 8 (Use a separate Writing Booklet)

Marks

- a) AB is a fixed chord of a circle. P is any point on the major arc.
The bisectors of $\angle PAB$ and $\angle PBA$ meet the circle at X and Y respectively.

4



- i) Copy the diagram into your Writing Booklet, showing the information given.
ii) Prove that XY is constant.
- b) A particle is projected from origin O with speed V m/s and angle θ to the horizontal.
- i) Show that the cartesian equation of its path is given by
- $$y = x \tan \theta - \frac{g \sec^2 \theta}{2V^2} x^2.$$
- ii) Prove that there are two paths possible through a point (a, h) on the path if
- $$(V^2 - gh)^2 > g^2(a^2 + h^2).$$
- c) Using mathematical induction, show that for each positive integer n there are
- unique positive integers p_n and q_n such that $(1 + \sqrt{2})^n = p_n + q_n \sqrt{2}$.

4

4

END OF EXAMINATION

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

St. Catherine's School

Yr. 12 4U. Trial HSC Solutions

Aug. 1997

Seet. A.

a) (i) $\int \sin 2x \cos x \, dx = \int 2 \sin x \cos x \cos x \, dx$
 $= \int 2 \sin x \cos^2 x \, dx$ let $u = \cos x$
 $du = -\sin x \, dx$
 $= 2 \int -u^2 \, du$
 $= -\frac{2}{3} u^3 + C = -\frac{2}{3} \cos^3 x + C$ (2)

(ii) $\int \sin 2x \cos 2x \, dx$
 $= \int \frac{1}{2} \sin 4x \, dx = -\frac{1}{8} \cos 4x + C$

— Too complicated
— look for
an easy way
out!

b) $\frac{2t^2}{(t-1)(t-2)(t-3)} = \frac{A}{t-1} + \frac{B}{t-2} + \frac{C}{t-3}$
 $2t^2 = A(t-2)(t-3) + B(t-1)(t-3) + C(t-1)(t-2)$
1, $2 = A \cdot 1 \cdot -2$ ie $A = 1$
2, $8 = B \cdot 1 \cdot -1$ ie $B = -8$
3, $18 = C \cdot 2 \cdot 1$ ie $C = 9$

A lot longer
this method
made it very
difficult for
some students.

$I = \int_4^5 \left(\frac{1}{t-1} - \frac{8}{t-2} + \frac{9}{t-3} \right) dt$
 $= \left[\ln(t-1) - 8 \ln(t-2) + 9 \ln(t-3) \right]_4^5$
 $= \left[\ln \frac{(t-1)(t-3)^9}{(t-2)^8} \right]_4^5$
 $= \ln \frac{4 \times 2^9}{3^8} - \ln \frac{3 \times 1}{2^8}$
 $= \ln \frac{2^{11}}{3^8} \times \frac{2^8}{3} = 19 \ln 2 - 9 \ln 3$ (5)

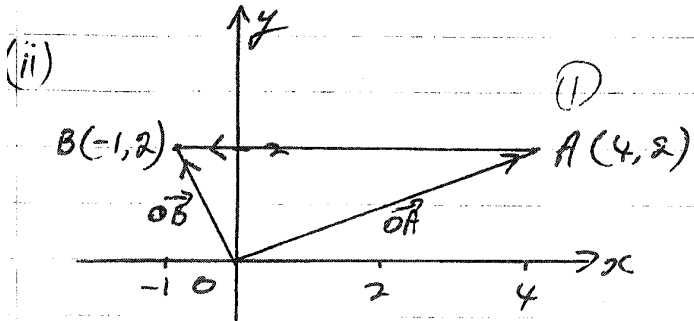
$$(a)(i) z_1 z_2 = (1+3i)(1-i) = 1-i+3i+3$$

$$= 4+2i \quad (1)$$

$$\frac{z_1}{z_2} = \frac{1+3i}{1-i} \times \frac{1+i}{1+i}$$

$$= \frac{1+4i-3}{2}$$

$$= \frac{-2+4i}{2} = -1+2i \quad (1)$$



(iii) $\frac{z_1}{z_2} - z_1 z_2 = \vec{AB}$

$$= -1-4$$

$$= -5$$

which is real.

Also, as from diagram (see No. 3600)

(b) (i) $z = \sqrt{3} - i$

$$= 2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = 2\left[\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right]$$

(ii) $z^5 = 2^5 \left[\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right]$

$$= 32\left[-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right] = -16(\sqrt{3} + i)$$

(iii) $\bar{z} = 2\left[\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right]$

$$(\bar{z})^5 = 2^5 \left[\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right] = 16(-\sqrt{3} + i)$$

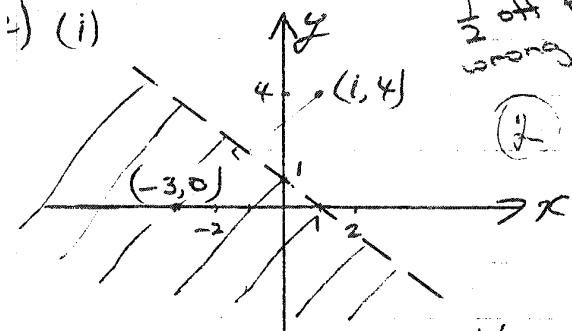
(iv) $\frac{z^5}{(\bar{z})^5} = \frac{16\left(\cos\frac{-5\pi}{6} + i\sin\frac{-5\pi}{6}\right)}{16\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)}$

$$= \cos\left(\frac{-10\pi}{6}\right) + i\sin\left(\frac{-10\pi}{6}\right)$$

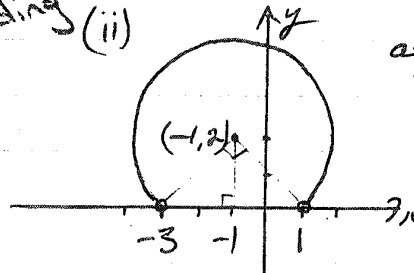
$$= \cos\frac{\pi}{3} + i\sin\frac{\pi}{3} = \frac{1}{2} + i\frac{\sqrt{3}}{2} \quad (1)$$

$\text{OR } \frac{-16(\sqrt{3} + i)}{16(-\sqrt{3} + i)}$

(i) $\frac{1}{2}$ off for wrong shading (ii)



$$|z - (-3)| < |z - (1+4i)|$$



$$\arg(2-i) - \arg(2+i) = \frac{\pi}{4}$$

$$\angle \text{ at circum.} = \frac{\pi}{4}$$

$$\angle \text{ at centre} = \frac{\pi}{2}$$

$$\text{Centre } (-1, 2)$$

$$2r^2 = 4^2$$

$$r^2 = 8$$

$$r = \sqrt{8}$$

$$\begin{aligned}
 \text{c) } I &= \int e^x \sin 2x \, dx \quad \text{let } u = \sin 2x \quad \frac{du}{dx} = 2 \cos 2x \quad v = e^x \\
 &= uv - \int v \cdot du \\
 &= e^x \sin 2x - \int e^x \cdot 2 \cos 2x \, dx \\
 &= e^x \sin 2x - 2 \left(e^x \cos 2x - \int e^x \cdot 2 \sin 2x \, dx \right) \quad \left\{ \begin{array}{l} \text{let } u_1 = \cos 2x \quad \frac{du_1}{dx} = -2 \sin 2x \\ v_1 = e^x \end{array} \right. \\
 &= e^x \sin 2x - 2e^x \cos 2x + 4 \int e^x \sin 2x \, dx \\
 2I &= e^x \sin 2x - 2e^x \cos 2x - 4I \\
 5I &= e^x \sin 2x - 2e^x \cos 2x \\
 I &= \frac{e^x \sin 2x - 2e^x \cos 2x}{5} = \frac{e^x}{5} (\sin 2x - 2 \cos 2x) + C
 \end{aligned}$$

$$\text{d) } \int_0^m \frac{(m-x)^2}{m^2+x^2} \, dx = \int_0^m \frac{m^2 - 2mx + x^2}{m^2+x^2} \, dx$$

$$= \int_0^m \left(1 - \frac{2mx}{m^2+x^2} \right) \, dx$$

$$= \left[x - m \ln(m^2+x^2) \right]_0^m$$

$$= m - m \ln(2m^2) - (0 - m \ln m^2)$$

$$= m - m \ln(2m^2) + m \ln m^2$$

$$= m \left(1 + \ln \left(\frac{m^2}{2m^2} \right) \right)$$

$$= m \left(1 + \ln \frac{1}{2} \right) = \underline{m(1 - \ln 2)} \quad \# \text{ (7)}$$

$$(a)(i) P(z) = 2z^3 - 5z^2 - 2z + 15$$

$$\begin{aligned} P(2+i) &= 2(2+i)^3 - 5(2+i)^2 - 2(2+i) + 15 \\ &= 2(8+12i-6-i) - 5(4+4i-1) - 2(2+i) + 15 \\ &= 4+22i-15-20i-4-2i+15 = 0 \end{aligned}$$

$\therefore (2+i)$ is a root of $P(z) = 0 \neq$ (1)

(ii) Since coeff. of $P(z)$ are integers, $2-i$ is also a root of $P(z) = 0$

$$(z-2-i)(z-2+i) = z^2 - 4z + 5 \quad \text{See No 56}$$

By division, $P(z) = (z^2 - 4z + 5)(2z + 3)$

Roots are $2+i, 2-i, -\frac{3}{2}$ (2)

$$b)(i) x^3 - px^2 + qx - r = 0 \quad \therefore \alpha + \beta + \gamma = p$$

Let $y = p - x \quad \therefore x = p - y$

Eqn. with roots $\alpha + \beta, \beta + \gamma, \gamma + \alpha$ is

$$(p-y)^3 - p(p-y)^2 + q(p-y) - r = 0$$

$$p^3 - 3p^2y + 3py^2 - y^3 - p(p^2 - 2py + y^2) + pq - qy - r = 0$$

$$-p^2y + 2py^2 - y^3 + pq - qy - r = 0$$

$$\frac{y^3 - 2py^2 + (p^2 + q)y - pq + r = 0}{x^3 - 2px^2 + (p^2 + q)x - pq + r = 0} \quad (3)$$

OR $x^3 - 2px^2 + (p^2 + q)x - pq + r = 0$

(ii) Product of roots $= (\alpha + \beta)(\beta + \gamma)(\gamma + \alpha) = \underline{pq - r}$ (4)

2)(i) Let $P(x) = (x-a)^m \cdot Q(x)$

then $P'(x) = m(x-a)^{m-1} \cdot Q(x) + (x-a)^m \cdot Q'(x)$

$$= (x-a)^{m-1} \cdot R(x)$$

which has multiplicity $(m-1) \neq$ (5)

(ii) Let $P(x) = x^4 - x^3 - 9x^2 - 11x - 4 = 0$

$$P'(x) = 4x^3 - 3x^2 - 18x - 11 = 0$$

$$P''(x) = 12x^2 - 6x - 18 = 0$$

$$\text{is } 6(x+1)(2x-3) = 0$$

Check for $P(-1) + P(\frac{3}{2})$. $P(-1) = 0$, $P(\frac{3}{2}) \neq 0$

$P(4) = 0$ by inspection.

Roots are $-1, -1, -1, 4$ (3)

(a) (i) $(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$ /

Very well done!

$$\begin{aligned} \text{Also } (\cos \theta + i \sin \theta)^5 &= \cos^5 \theta + 5 \cos^4 \theta \cdot i \sin \theta + 10 \cos^3 \theta \cdot i^2 \sin^2 \theta \\ &\quad + 10 \cos^2 \theta \cdot i^3 \sin^3 \theta + 5 \cos \theta \cdot i^4 \sin^4 \theta + i^5 \sin^5 \theta \\ &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta + (5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta) \end{aligned}$$

$$\begin{aligned} \cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta (1 - 2 \cos^2 \theta + \cos^4 \theta) \\ &= 11 \cos^5 \theta - 10 \cos^3 \theta + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta \\ \cos 5\theta &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \quad (3) \end{aligned}$$

(ii) Let $x = \cos \theta$. $\cos 5\theta = 16x^5 - 20x^3 + 5x$
 $= x(16x^4 - 20x^2 + 5)$

When $\cos 5\theta = 0$ then $x(16x^4 - 20x^2 + 5) = 0$

$$\therefore 5\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}$$

$$\text{i.e. } \theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{\pi}{2}, \frac{7\pi}{10}, \frac{9\pi}{10}$$

When $x = \cos \frac{\pi}{2}$, solution to $x = 0$

$\therefore$ Roots of $16x^4 - 20x^2 + 5 = 0$ are

$$x = \cos \frac{\pi}{10}, \cos \frac{3\pi}{10}, \cos \frac{7\pi}{10}, \cos \frac{9\pi}{10} \quad (3)$$

7) (i) $I_n = \int \frac{x^n}{\sqrt{x^2 - a^2}} dx$

$$= \int x^{n-1} \cdot \frac{x}{\sqrt{x^2 - a^2}} dx = \int x^{n-1} \frac{d}{dx} (\sqrt{x^2 - a^2}) dx$$

$$= x^{n-1} \sqrt{x^2 - a^2} - \int \sqrt{x^2 - a^2} \cdot (n-1) x^{n-2} dx$$

$$= x^{n-1} \sqrt{x^2 - a^2} - (n-1) \left[\int \frac{x^n}{\sqrt{x^2 - a^2}} - \int \frac{a^2 x^{n-1}}{\sqrt{x^2 - a^2}} \right]$$

$$I_n = x^{n-1} \sqrt{x^2 - a^2} - (n-1) I_n + a^2 (n-1) I_{n-2}$$

$$n \cdot I_n - (n-1) a^2 \cdot I_{n-2} = x^{n-1} \sqrt{x^2 - a^2} \quad (3)$$

$$(b) (ii) \int_2^4 \frac{x^4}{\sqrt{x^2-4}} dx = ? \quad n=4, a=2$$

$$-4): 4I_4 - 3 \cdot 4 \cdot I_2 = \left[x^3 \sqrt{x^2-4} \right]_2^4 = 64\sqrt{12} = 128\sqrt{3}$$

$$-2): 2I_2 - 4I_0 = \left[x \sqrt{x^2-4} \right]_2^4 = 4\sqrt{12} = 8\sqrt{3}$$

$$\begin{aligned} I_0 &= \int_2^4 \frac{1}{\sqrt{x^2-4}} dx = \left[\ln(x + \sqrt{x^2-4}) \right]_2^4 \\ &= \ln\left(\frac{4+\sqrt{12}}{2}\right) \\ &= \ln(2+\sqrt{3}) \end{aligned}$$

$$\therefore 2I_2 = 4\ln(2+\sqrt{3}) + 8\sqrt{3}$$

$$\begin{aligned} 4I_4 &= 6 \cdot 2I_2 + 128\sqrt{3} \\ &= 24\ln(2+\sqrt{3}) + 48\sqrt{3} + 128\sqrt{3} \\ &\quad 24\ln(2+\sqrt{3}) + 176\sqrt{3} \end{aligned}$$

$$\therefore I_4 = 6\ln(2+\sqrt{3}) + 44\sqrt{3} \quad \textcircled{=}$$

Section B. 40. Trial HSC 1997.

5(a) $y = \frac{x^2}{x^3+4}$

i) Vertical asymptote $x^3+4=0$ is $x = -\sqrt[3]{4}$ (≈ -1.6)
 Horizontal " $x \rightarrow \infty, y = \frac{x^2}{\frac{x^3}{x^3} + \frac{4}{x^3}} = 0$ i.e. $y = 0$

ii) $\frac{dy}{dx} = \frac{(x^3+4)2x - x^2 \cdot 3x^2}{(x^3+4)^2}$
 $= \frac{2x^4 + 8x - 3x^4}{(x^3+4)^2} = \frac{8x - x^4}{(x^3+4)^2}$ (1)

S.P. $8x - x^4 = 0$
 $x(8 - x^3) = 0 \Rightarrow x = 0, 2$

When $x = 0, y = 0$

When $x = 2, y = \frac{1}{3}$

$x = 0^-, \frac{dy}{dx} = -ve$

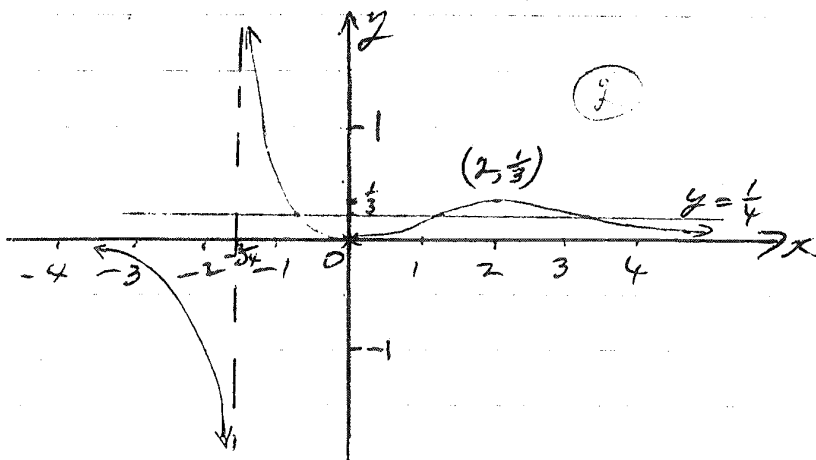
$x = 2^-, \frac{dy}{dx} = +ve$

$x = 0^+, \frac{dy}{dx} = +ve$

$x = 2^+, \frac{dy}{dx} = -ve$

$(0,0)$ is Min. T.P.

$(2, \frac{1}{3})$ is Max. T.P. (2)

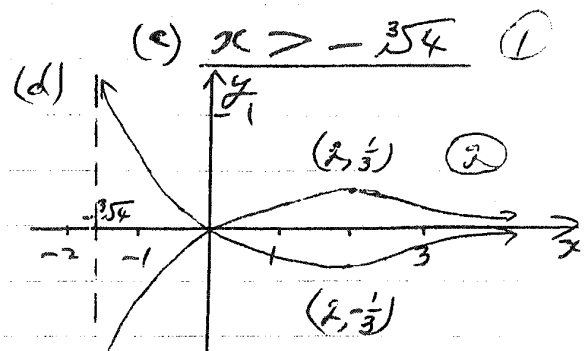
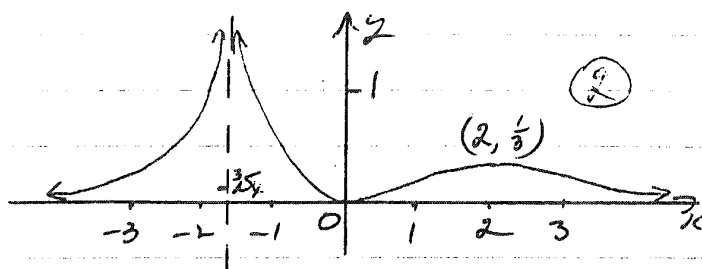


(iv) $4x^2 = x^3 + 4$

$\frac{x^2}{x^3+4} = \frac{1}{4}$

Draw $y = \frac{1}{4}$

It cuts the graph in 3 places. (1)



$$\begin{aligned}
 6. (a) \int \cos^2 y \, dy &= \frac{1}{2} \int (1 + \cos 2y) \, dy \\
 &= \frac{1}{2} \left(y + \frac{1}{2} \sin 2y \right) + C \\
 &= \underline{\frac{1}{2}y + \frac{1}{4} \sin 2y + C} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 b) (i) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} &= 1 \quad \text{Asymptotes } y = \pm \frac{b}{a}x \\
 \therefore \frac{x^2}{a^2} - \frac{y^2}{a^2} &= 1 \quad \frac{b}{a} = 1 \quad \therefore b^2 = a^2
 \end{aligned}$$

$$\begin{aligned}
 x(3, 2): \quad \frac{9}{a^2} - \frac{4}{a^2} &= 1 \quad \Rightarrow 9 - 4 = a^2 \quad \therefore a^2 = 5 \\
 \text{Eqn. } \frac{x^2}{5} - \frac{y^2}{5} &= 1 \quad (2) \\
 \text{OR } x^2 - y^2 &= 5
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad b^2 &= a^2(e^2 - 1) \\
 5 &= 5(e^2 - 1) \quad \therefore e^2 - 1 = 1 \\
 e^2 &= 2, \quad e = \sqrt{2} \quad (1)
 \end{aligned}$$

$$\text{Foci: } (\pm a e, 0), \quad (\pm \sqrt{5} \cdot \sqrt{2}, 0) \quad \underline{(\pm \sqrt{10}, 0)} \quad (2)$$

$$\begin{aligned}
 c) \quad \frac{x^2}{27} + \frac{y^2}{9} &= 1 \quad \perp x + y = 10 \Rightarrow \text{grad} = 1 \\
 \text{Subst. } y &= x + c \text{ in eqn.}
 \end{aligned}$$

$$\frac{x^2}{27} + \frac{x^2 + 2xc + c^2}{9} = 1$$

$$x^2 + 3x^2 + 6xc + 3c^2 = 27$$

$$4x^2 + 6xc + 3c^2 - 27 = 0$$

$$\therefore 36c^2 - 16(3c^2 - 27) = 0$$

$$9c^2 - 4(3c^2 - 27) = 0$$

$$9c^2 - 12c^2 - 108 = 0$$

$$\Rightarrow 3c^2 = 108$$

$$c^2 = 36$$

$$c = \pm 6$$

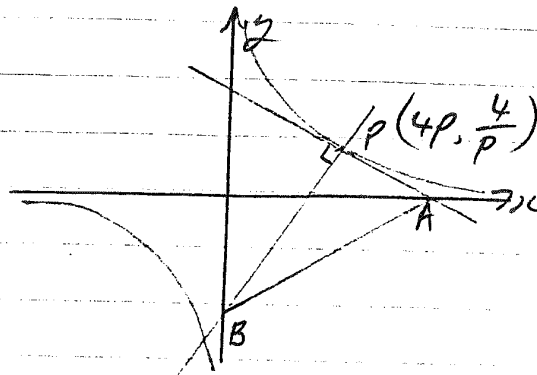
$$\text{Tangents are } y = x + 6 \quad |$$

$$\underline{y = x - 6} \quad (3)$$

c) (i) $y = \frac{16}{x}$

$$\frac{dy}{dx} = -\frac{16}{x^2}$$

At P , $\frac{dy}{dx} = -\frac{1}{p^2}$



of tangent:

$$-\frac{1}{p^2} = \frac{y - \frac{4}{p}}{x - 4p}$$

$$-x + 4p = p^2 y - 4p$$

$$x + p^2 y = 8p \quad (1)$$

Eqn. of normal:

$$p^2 = \frac{y - \frac{4}{p}}{x - 4p}$$

$$p^2 x - 4p^3 = y - \frac{4}{p}$$

$$p^3 x - 4p^4 = py - 4$$

$$p^3 x - py = 4(p^4 - 1) \quad (2)$$

At A, $y = 0 \therefore x = 8p$

At B, $x = 0 \therefore y = \frac{4}{p}(1 - p^4)$

$$\begin{aligned} \text{Dist. AP} &= \sqrt{(4p)^2 + \left(\frac{4}{p}\right)^2} \\ &= \sqrt{16p^2 + \frac{16}{p^2}} \\ &= 4\sqrt{\frac{p^4 + 1}{p^2}} \end{aligned}$$

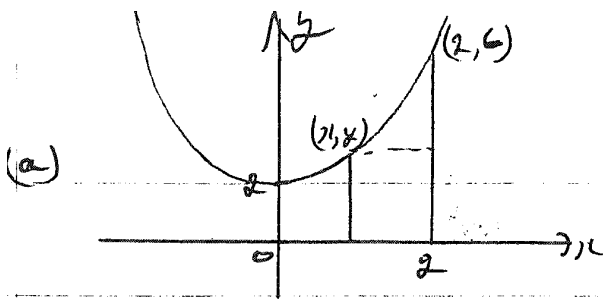
$$\begin{aligned} \text{Dist. PB} &= \sqrt{(4p)^2 + (4p^3)^2} \\ &= \sqrt{16p^2 + 16p^6} \\ &= 4\sqrt{p^2 + p^6} \end{aligned}$$

$$\text{Area } \triangle PAB = \frac{1}{2} \cdot 4\sqrt{\frac{p^4 + 1}{p^2}} \cdot 4\sqrt{p^2 + p^6}$$

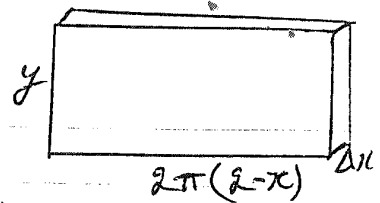
$$= 8\sqrt{(p^4 + 1)(p^3 + 1)}$$

sq. units.

(9)



$$\Delta A = (2-x) \cdot y$$



$$\Delta V = 2\pi(2-x)y \Delta x$$

$$Vol. = 2\pi \int_0^2 (2-x)(x^2+2) dx$$

$$= 2\pi \int_0^2 (2x^2 + 4 - x^3 - 2x) dx$$

$$= 2\pi \left[\frac{2}{3}x^3 + 4x - \frac{1}{4}x^4 - x^2 \right]_0^2$$

$$= 2\pi \left\{ \left(\frac{16}{3} + 8 - 4 - 4 \right) - (0) \right\} = \frac{32\pi}{3} m^3$$

(4)

5) (i) $\triangle ADE \parallel \triangle ABC$ in ratio 1:2

$$\therefore \frac{2h}{3} = \frac{4-5}{4} = \frac{2-5}{2}$$

$$8h = 12 - 35$$

$$h = \frac{3}{2} \left(1 - \frac{1}{4}5 \right) \neq \textcircled{3}$$

(ii) $\Delta A = \frac{1}{3}h$

$$= \frac{1}{3} \cdot \frac{3}{2} \left(1 - \frac{1}{4}5 \right)$$

$$\Delta A = \frac{1}{2} \left(1 - \frac{1}{4}5 \right) \neq \textcircled{2}$$

ii) $\Delta V = \frac{1}{2} \left(1 - \frac{1}{4}5 \right) \Delta s$

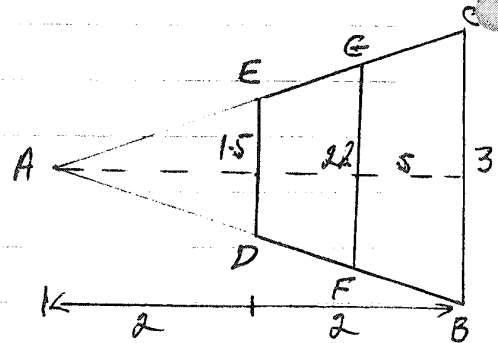
$$V = \lim_{\Delta s \rightarrow 0} \sum_{s=0}^2 \frac{1}{2} \left(1 - \frac{1}{4}s \right) \Delta s$$

$$= \frac{1}{2} \int_0^2 \left(1 - \frac{1}{4}s \right) ds$$

$$= \frac{1}{2} \left[s - \frac{1}{8}s^2 \right]_0^2$$

$$= \frac{1}{2} \left\{ \left(2 - \frac{4}{8} \right) - (0) \right\}$$

$$V = \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4} m^3 \textcircled{3}$$



(1) $EA = 1.5$

$2.2, \frac{2}{3} = 2.2 \times 3 = 6.6$

$1.5 \times 3 = 4.5$

$2.2 \times 3 = 6.6$

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$1.5 \times 3 = 4.5$

(a) Let $\angle PAB = 2\alpha$

$\angle PBA = 2\beta$

Since AB is fixed (constant length),
 $\angle APB = h$ will be constant for
 varying positions of P, since angles at cir.
 ending on the same chord are equal.

PX is a chord.

$$\angle PAB = \angle PBX = \alpha$$

$$\text{In } \triangle PAB, h + 2\alpha + 2\beta = 180^\circ$$

$$h + 2(\alpha + \beta) = 180^\circ$$

But $\angle XBY = \alpha + \beta$ is constant (angle at circum. subtended by XY)
 A constant angle, will be subtended by a chord
 that does not change in length. $\therefore XY$ is constant (4)

Derive

$$(i) \begin{cases} x = V \cos \theta t & \text{ie. } t = \frac{x}{V \cos \theta} \\ y = V \sin \theta t - \frac{g t^2}{2} \end{cases}$$

$$\therefore y = \frac{V \sin \theta \cdot x}{V \cos \theta} - \frac{g}{2} \frac{x^2}{V^2 \cos^2 \theta}$$

$$\therefore y = x \tan \theta - \frac{g x^2 \sec^2 \theta}{2 V^2} \quad \# \quad (2)$$

$$(i) \text{ At } (a, h): h = a \tan \theta - \frac{g(1 + \tan^2 \theta) a^2}{2 V^2}$$

$$\tan^2 \theta \left(\frac{g a^2}{2 V^2} \right) - a \tan \theta + h + \frac{g a^2}{2 V^2} = 0$$

$$\Delta > 0 \text{ for 2 values: } a^2 > \frac{4 g a^2}{2 V^2} \left(h + \frac{g a^2}{2 V^2} \right)$$

$$V^4 - 2 g h V^2 > g^2 a^2$$

$$(V^2 - g h)^2 > g^2 (a^2 + h^2) \quad \# \quad (2)$$

$$\text{When } n=1, \text{ LHS} = (1 + \sqrt{2})^1 = 1 + \sqrt{2} \quad \text{RHS} = p_1 + q_1 \sqrt{2} = \text{LHS}$$

when $p_1 = 1, q_1 = 1$. Thus unique integers p_i, q_i can be found.

$$\text{use } (1 + \sqrt{2})^h = p_h + q_h \sqrt{2}$$

$$= p_{h+1} + q_{h+1} \sqrt{2} \quad (+)$$

$$\begin{aligned} (\sqrt{2})^{h+1} &= (1 + \sqrt{2})^1 (1 + \sqrt{2})^h \\ &= (1 + \sqrt{2}) (p_h + q_h \sqrt{2}) \end{aligned}$$

Since p_h, q_h are unique, so are p_{h+1}, q_{h+1} .

(3) Since it is true for $n=1$, then it is true